

IFAC World Congress 2026

RoboRacer Autonomous Racing

Technical Report

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IFAC RoboRacer Team Sookshark

IFAC 2026 RoboRacer

Vehicle Hardware Specification

Competition: 29th Roboracer Autonomous Racing Competition

Team Name: Sookshark

Vehicle: Traxxas Slash 4x4-based RoboRacer

Hardware Specification

Category	Specification	Note
Chassis	Traxxas Slash 4x4	1EA
LiDAR	Livox MID-360	1EA
Onboard computer	MeLE Overclock 4C	1EA
ESC	Makerbase MKS ESC MINI V6.7 Pro	Final model
Battery	Traxxas Power Cell 5000 mAh 3-cell LiPo	1EA
Power Converter	DC 8V-40V to DC 12V 10A 120W stabilizer converter	1EA

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1. Introduction

This report describes the technical configuration of the vehicle for the 29th Roboracer Autonomous Racing Competition at IFAC 2026, held at BEXCO in Busan from August 24 to August 27, 2026. IFAC 2026 RoboRacer is an in-person autonomous racing competition in which each team brings a physical 1:10 scale autonomous vehicle and its own software stack.

The team built the vehicle on a Traxxas Slash 4x4 1:10 scale RC platform and integrated a Livox MID-360 LiDAR, MeLE Overclock 4C onboard computer, Makerbase MKS ESC MINI V6.7 Pro, and a ROS 2 Humble-based perception/planning/control stack. The development goal is to implement a robust autonomous racing stack including Livox/IMU-based FAST-LIO 3D mapping, 2D occupancy map generation, fixed-map localization, raceline tracking, obstacle avoidance, and safety stop behavior.

2. Hardware and Platform Design

A. Vehicle Platform

Category	Specification
Platform	Traxxas Slash 4x4
Vehicle Type	1/10 scale RC-based RoboRacer/F1TENTH vehicle
Onboard Computer	MeLE Overclock 4C
LiDAR	Livox MID-360
ESC	Makerbase MKS ESC MINI V6.7 Pro

The Traxxas Slash 4x4 is a 1:10 scale four-wheel-drive RC platform with front-wheel steering. The vehicle preserves the base mechanical structure of the RC

platform while mounting the LiDAR and onboard computer on the upper frame to form an autonomous racing platform.

B. Power and Electrical System

Category	Specification
Battery	Traxxas Power Cell 5000 mAh, 3-cell LiPo
Power Converter	DC 8V-40V to DC 12V 10A 120W stabilizer converter
Main Connectors	XT60, Traxxas/TRX adapter cable, XT60 Y-splitter cable
Wiring Parts	DC 5.5 mm two-wire cable, heat-shrink tube, jumper cable

The main power source is one Traxxas Power Cell 5000 mAh 3S LiPo battery. Battery power is distributed to the ESC and auxiliary power system through XT60 and Traxxas/TRX adapter cables. A DC 12 V 10 A stabilizer converter is used to provide stable power to the Livox MID-360 and the onboard computer.

C. Sensor Mounting Design

The Livox MID-360 is mounted on the upper part of the vehicle to observe the surrounding environment in 360 degrees. In the current perception configuration, the LiDAR is set at approximately 0.27 m forward and 0.18 m above the base_link frame, with a mounting yaw correction of approximately +90 degrees relative to the vehicle. This mounting transform is managed consistently in both the ROS 2 static transform and the scan projection configuration.

3. Sensors and Sensor Processing Algorithms

A. Livox MID-360

Category	Specification
Sensor	Livox MID-360
ROS Input	/livox/lidar (livox_ros_driver2/CustomMsg)
ROS Outputs	/scan, /livox/points_raw, /livox/points_deskewed, /cloud_registered, /Odometry, /fixed_map
Purpose	FAST-LIO 3D mapping, 2D occupancy map generation, fixed-map localization, obstacle perception

The Livox MID-360 publishes data in ROS 2 as CustomMsg messages on the /livox/lidar topic. In the real-time driving path, the system directly converts CustomMsg data into LaserScan messages to generate /scan. This reduces the cost of intermediate point-cloud conversion and supports real-time processing on the compact onboard computer.

Item	Value
target frame	base_link
sensor x / y / z	0.27 m / 0.0 m / 0.18 m
sensor yaw	1.57079632679 rad
height filter	-0.15 m to 0.35 m
range filter	0.2 m to 10.0 m
angle increment	0.00436 rad

B. IMU Yaw-Rate-Based Deskew

A Livox point cloud contains points acquired at different times within a single scan. If the vehicle rotates while the scan is being acquired, treating all points as if they were captured at one instant can bend walls or distort corners. The system performs planar motion compensation using each point's `offset_time`, odometry-based linear velocity, and the Livox IMU yaw rate.

Deskew validation compares `/livox/points_raw` and `/livox/points_deskewed` from the same rosbag. The validation criterion is not merely whether point locations change, but whether wall straightness and corner geometry become closer to the real environment.

C. 3D Clustering

3D clustering is a perception module that separates obstacle candidates from the LiDAR point cloud and estimates each obstacle's position and size. The default input is `/livox/points_deskewed`, and the processing sequence consists of ROI filtering, downsampling, Euclidean clustering, and bounding box computation.

Step	Description
ROI Filtering	Limits the valid driving region around the vehicle and constrains the z-axis range
Downsampling	Reduces computational load using voxel grid or point sampling
Euclidean Clustering	Groups nearby points into clusters based on 3D Euclidean distance
Bounding Box	Computes the center and x/y/z boundaries of each cluster
Driving Decision Interface	Classifies obstacle clusters that affect the forward driving path for avoidance or speed reduction

4. Algorithm Design

A. Environment Perception

The environment perception module uses both the 3D point cloud from the Livox MID-360 and the 2D scan projected from it. The 3D point cloud is used to estimate the spatial position and shape of obstacles, while the 2D scan is used for real-time drivable-space assessment. Based on these inputs, the system identifies track boundaries, frontal obstacles, and available driving space, and forwards relevant obstacle information to the local planner and safety module when it affects the vehicle path.

B. FAST-LIO 3D-2D Fusion Mapping and Localization

The perception mapping pipeline is centered on FAST-LIO-based LiDAR-inertial odometry, which combines the Livox MID-360 point cloud with IMU measurements. FAST-LIO uses IMU-predicted states to deskew each LiDAR scan and perform scan-to-map updates, producing a continuous 3D trajectory and registered point cloud. This 3D map preserves surrounding structural geometry and provides richer geometric information than a simple 2D scan projection, especially during rotations, cornering, and in environments with height structure.

3D-2D fusion mapping is performed using the map-frame 3D point cloud generated by FAST-LIO. The 3D map is used as the fixed PCD map and localization reference, while a 2D occupancy map is generated from the same coordinate frame by applying height filtering and floor alignment. Planning and safety use the computationally lighter 2D map and /scan, while localization and map-quality validation use the 3D structure and FAST-LIO odometry.

The core system structure maintains FAST-LIO native mapping, fixed-map PCD localization, and a 2D occupancy map derived from the 3D map in the same map frame. This combines 3D structure-based localization and 2D path planning in one coordinate frame: the fixed PCD map is used for localization, while the 2D occupancy map and /scan are used for planning and safety.

C. FAST-LIO Fixed-Map Localization

Racing requires both a reliable initial pose in the map frame before driving and continuous pose tracking during motion. The system uses static global localization to estimate the initial pose while the vehicle is stationary, then uses FAST-LIO fixed-map localization to perform LiDAR/IMU-based scan-to-map updates against the prepared 3D PCD map.

Module	Role	Input/Output
Static global localization	Estimates the initial pose in the 3D map while the vehicle is stationary	Input: /livox/points_deskewed, fixed 3D map / Output: /initialpose, localization status
FAST-LIO fixed-map localization	Tracks LiDAR/IMU scan-to-map alignment against an immutable PCD map	Input: /livox/lidar, /livox/imu, /initialpose, fixed_map.pcd / Output: /Odometry, /cloud_registered, TF, /fast_lio/localization_status
3D-2D map fusion	Maintains the 3D PCD map and 2D occupancy map in the same map frame	Output: /fixed_map, occupancy grid, map frame for path planning

FAST-LIO fixed-map localization loads a prebuilt immutable PCD map into the ikd-tree and uses live MID-360 scans and IMU measurements to perform ESIKF scan-to-plane updates. In localization mode, map insertion, FOV deletion, PCD saving, and cumulative /Laser_map publication are disabled so that the map does not change during driving. Outputs include map-frame /Odometry, /cloud_registered, TF, and localization status.

Static global localization accumulates /livox/points_deskewed for several seconds while the vehicle is stationary, then computes an initial map-frame pose

using Open3D-based global registration and ICP. This pose is passed to FAST-LIO fixed-map localization through `/initialpose`. During driving, FAST-LIO uses IMU prediction and fixed-map scan matching, with feature and innovation gates to prevent arbitrary pose jumps.

The 2D map is derived from the FAST-LIO 3D map and is used for planning and RViz monitoring. The 3D fixed map and 2D occupancy map share the same source-to-map transform, so the global seed, FAST-LIO localization, and path planning all operate in the same map frame.

Validation Item	Criterion
Initial Registration Quality	Evaluate static global localization score, ICP fitness, and RMSE
FAST-LIO Tracking State	<code>/fast_lio/localization_status</code> remains TRACKING with zero non-monotonic stamps
Pose Continuity	Translation/yaw steps after scan-to-map updates remain within vehicle dynamics
3D-2D Frame Consistency	Fixed PCD map, occupancy grid, <code>/Odometry</code> , and <code>/path</code> align in the same map frame
Resource Budget	Maintain LiDAR 10 Hz processing and localization output on the onboard computer

D. Path Planning

The global path is the reference path that the vehicle follows in the map frame. The system creates track centerline waypoints on the 2D occupancy map derived from the FAST-LIO 3D map, then improves them into a curvature-constrained raceline considering track width and curvature.

The local planner selects a short-horizon drivable path using /scan, FAST-LIO odometry, and 3D cluster information. It uses a Follow-the-Gap/Pure Pursuit structure that is robust to track-boundary gaps and LiDAR noise, and generates pass-by trajectories when another vehicle or obstacle cluster is detected.

Planning Method	Role
Global Path	Generate map-frame waypoints, centerline, and raceline
Follow-the-Gap	Select the direction of the widest free gap from scan data
DWA Variant	Evaluate speed/steering trajectory candidates using a cost function
Pure Pursuit-Assisted Avoidance	Track the global path or pass-by waypoints

E. Motion Planning and Vehicle Control

The Traxxas Slash 4x4 is an Ackermann-type vehicle platform with front-wheel steering. Therefore, the control input is separated into steering angle and drive speed. Lateral control uses Pure Pursuit or a Stanley controller, while longitudinal control uses P/PD control based on VESC/odometry speed feedback.

Pure Pursuit selects a target point at a lookahead distance ahead of the current vehicle position and computes the curvature required to reach that point. The curvature is converted to a steering angle using $\text{steering_angle} = \text{atan}(\text{wheelbase} * \text{curvature})$. The lookahead distance increases with vehicle speed to smooth steering behavior at higher speeds.

The Stanley controller can be used when the path centerline and vehicle heading are estimated reliably. It combines heading error and cross-track error. Since the cross-track error term can become large at low speeds, steering angle saturation and steering-rate limits must be applied.

Longitudinal control reduces the error between the target speed and the current speed. The target_speed generated by the planner is compared with the current_speed from VESC/odometry, and a P/PD controller is applied in the form $\text{throttle_cmd} = k_p * \text{speed_error} + k_d * d(\text{speed_error})/dt$.

For driving stability, the system limits maximum steering angle, steering-rate change, maximum speed, acceleration and deceleration rate, and speed based on obstacle distance. Speed is increased on straight segments and reduced in high-curvature sections. The speed is also limited according to the distance to other vehicles, available pass-by space, and track-boundary gaps. If /safety/brake is true, the emergency stop command takes priority over normal planner and controller commands.

5. Validation Plan

Validation Item	Description
Sensor and TF	Check Livox connection, /scan orientation, and base_link -> livox_frame transform
Deskew	Compare wall straightness between raw and deskewed clouds
Mapping	Compare FAST-LIO 3D map, 2D occupancy map, and fixed-map localization overlay
Clustering	Validate 3D cluster center points and bounding box accuracy
Control	Validate low-speed waypoint tracking, obstacle slowdown/stop behavior, and safety brake

6. Conclusion

The vehicle is a 1:10 scale autonomous racing system integrating a Traxxas Slash 4x4 platform, Livox MID-360 LiDAR, MeLE Overclock 4C, and Makerbase MKS ESC MINI V6.7 Pro for IFAC 2026 RoboRacer. The ROS 2 Humble-based perception pipeline processes Livox/IMU inputs through FAST-LIO and operates 3D mapping, 2D occupancy mapping, and fixed-map localization in a single map frame.

For perception and localization, the system uses FAST-LIO LiDAR-inertial odometry, fixed PCD map localization, 3D-2D fusion mapping, IMU yaw-rate-based deskew, 2D scan corridor obstacle detection, and 3D point cloud clustering. The 3D map is used for localization and structural validation, while the 2D occupancy map and /scan are used for planning and safety.

For control, the system applies Pure Pursuit or Stanley-based lateral control and P/PD-based longitudinal speed control matched to the Ackermann vehicle model. It also improves driving stability using speed limits based on frontal obstacles, TTC, steering angle, and localization stability, together with a safety brake.

7. References

[1] 29th Roboracer Autonomous Racing Competition at IFAC 2026, <https://2026ifac-roboracer.com/>

[2] IFAC 2026 RoboRacer Korean Page, <https://2026ifac-roboracer.com/kr/index.html>

[3] ROS 2 Humble Documentation

[4] Livox MID-360 User Manual

[5] FAST-LIO2 / FAST-LIO ROS2 Documentation

[6] Open3D Documentation

[7] F1TENTH / RoboRacer Autonomous Racing References

[8] VESC Project Documentation